



How To Use This Guide

This guide walks you through every stage of the IBDP Biology Internal Assessment. It is organised to match the official IB criteria, so you can use it from your very first planning session right through to your final submission check.

What is the IA?

The Internal Assessment is a scientific investigation worth 20% of your final IB Biology grade. It is marked by your teacher and then moderated by the IB. You have approximately 10 hours of class time, but most students spend considerably more time on it outside class.

Your IA is marked across four criteria:

Criterion	What it assesses
Research Design (RD)	How well you planned your investigation — question, variables, method
Data Analysis (DA)	How well you processed and presented your raw data
Conclusion (C)	How well you answered your research question using your data
Evaluation (E)	How honestly and specifically you assessed the strengths and weaknesses of your method

Before you start reading

Each criterion section includes: what the examiner is looking for, a checklist you can tick off, worked examples of strong and weak writing, and the most common mistakes that cost students marks.



Section 1: Research Design

Research Design is the foundation of your entire IA. A weak research question or poorly justified method will limit your score across all four criteria, not just this one.

1.1 The Research Question

Your research question (RQ) must be:

- Focused — it investigates one specific relationship
- Measurable — you can collect numerical data to answer it
- Biologically relevant — it connects to a concept from the IB Biology course
- Feasible — you can actually do it safely with available equipment

Strong vs Weak Research Questions

- ✗ Weak: “Does light affect plants?”
Too vague. What aspect of light? What aspect of plants? How will you measure it?
- ✓ Strong: “How does light intensity (100–500 lux, in steps of 100) affect the rate of photosynthesis in Elodea, as measured by oxygen bubble production per minute?”
Specific, measurable, and clearly scoped.

1.2 Variables

You must identify and explain three types of variables:

Variable Type	Definition	Example
Independent (IV)	What you deliberately change	Light intensity (100, 200, 300, 400, 500 lux)
Dependent (DV)	What you measure	Number of O ₂ bubbles produced per minute
Controlled (CV)	What you keep the same to ensure a fair test	Temperature, CO ₂ concentration, Elodea mass, water pH

Critical Rule: Operationalise your variables

Do not just name your variables. Explain HOW you will measure them, with UNITS.

- ✗ Weak: “Temperature will be controlled.”
- ✓ Strong: “Water temperature will be maintained at 25°C (±0.5°C) using a thermostatically controlled water bath, checked with a calibrated digital thermometer before each trial.”

1.3 Hypotheses

You must state both a null hypothesis (H₀) and an alternate hypothesis (H₁). These are not optional — they are required for a strong conclusion as well.